CS 473 (Spring 2025) Yiming Qin (yimingq3@illinois.edu)

Problem 9.1

Solution:

Reduction from 3-SAT

Given a 3CNF formula with n variables $x_1, \ldots, x_n$ and m clauses $C_1, \ldots, C_m$, where each clause C_j contains exactly three literals.

Construct a DAG $G = (V, E)$, a coloring function $c : E \to \{1, \ldots, m\}$, and vertices s and t such that the formula is satisfiable iff there exists a path from s to t in G that encounters at least $\ell = m$ distinct colors.

DAG Construction

- For each variable x_i, construct a gadget consisting of two paths from a vertex u_i to v_i:
  - The upper path P_i^T corresponds to assigning $x_i = \text{true}$.
  - The lower path P_i^F corresponds to assigning $x_i = \text{false}$.

  These paths form a "diamond" structure.

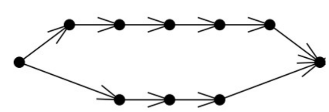

Figure 1: gadget

- Each clause C_j is assigned a unique color $j \in \{1, \ldots, m\}$.
- For each occurrence of literal x_i in clause C_j, insert an edge labeled with color j on the upper path P_i^T.
- For each occurrence of literal $\neg x_i$ in clause C_j, insert an edge labeled with color j on the lower path P_i^F.
- Connect the gadgets in sequence:

$$u_1 \to v_1 \to u_2 \to v_2 \to \cdots \to u_n \to v_n.$$

  Merge v_i and u_{i+1} ($i = 1, 2, \cdots, n-1$) into a singal node u_{i+1} (no colored path between them).

  Let the source vertex $s \leftarrow u_1$ and the sink vertex $t \leftarrow v_n$.

  Thus we have the DAG:

$$s \to u_2 \to \cdots \to u_n \to t.$$

  each $\to$ represents a gadget.

Correctness of the Reduction

($\Rightarrow$) Suppose the 3CNF formula is satisfiable. Then there exists a truth assignment such that every clause C_j is satisfied. Construct an s-t path in G as follows:

- For each variable x_i, if $x_i =$ true, follow the upper path P_i^T; otherwise, follow P_i^F.

Because the assignment satisfies all clauses, for each clause C_j, at least one literal in C_j is true under the assignment. Therefore, the corresponding path will include an edge colored with j. Hence, the path encounters all m colors.

($\Leftarrow$) Suppose there is an s–t path in G that encounters at least m distinct colors. Because each gadget has only two disjoint paths, and the entire graph is a sequence of such gadgets, the path must make exactly one choice (true/false) per variable. Define the truth assignment accordingly:

- If the path takes P_i^T, assign $x_i =$ true.
- If the path takes P_i^F, assign $x_i =$ false.

Since the path encounters all m colors, each clause color must appear on the path. Therefore, for each clause C_j, at least one of its literals is satisfied under this assignment. Thus, the 3CNF formula is satisfiable.

Complexity

The number of vertices and edges in G is $O(nm)$. The Colorful-Path problem is in NP, as the reduction can be done in polynomial time and a path can be verified in polynomial time.